

SEMESTER 2 EXAMINATION 2024/25

ADVANCED MACHINE LEARNING

Duration 120 mins (2 hours)

This paper is a WRITE-ON examination paper.

You **must** write your Student ID on this Page and must not write your name anywhere on the paper.

All answers should be written within the designated boxes in this examination paper and sufficient space is provided for each question.

If, for some reason, space is required to complete or correct an answer to a question, use the “Additional Space” provided on the facing or adjacent page to the question. Clearly indicate which question the answer corresponds to.

No credit will be given for answers presented elsewhere and without clear indication of to what question they correspond. Blue answer books may be used for scratch; they will be discarded without being looked at.

Answer all parts of the question in section A (40 marks) and ALL three questions from section B (20 marks each)

Student ID:

Question	Mark	Arithmetic checked	Double Marked
A1	/40		
B2	/20		
B3	/20		
B4	/20		
Total:	/100		

University approved calculators MAY be used.

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

9 page examination paper

Section A

A 1

- (a) In classification, what is the effect of increasing the number of training examples on generalisation performance? Explain why this is. [5 marks]

Increasing the number of training examples tends to improve generalisation performance. This is because each example tends to rule out spurious rules that might otherwise explain the classification of the training examples by chance, but does not reflect the true classification rule.

- (b) Briefly describe the random forest algorithm **and** explain why it is effective. [5 marks]

Random forest computes a large number of shallow decision trees where each tree uses a different subset of the features and is trained on a different bootstrap of the training examples. The final decision is made by voting for classification or averaging for regression. It is effective because each decision tree is fairly independent of the others and by averaging a large number of these we substantially reduce the variance, thus leading to a lower expected generalisation error.

- (c) Show that an empirical covariance matrix, $\mathbf{C}$, can be written as $\mathbf{X}\mathbf{X}^T$ and hence prove that its eigenvalues are non-negative. [5 marks]

The empirical covariance matrix is given by

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^T.$$

Define the matrix $\mathbf{X} = \frac{1}{\sqrt{m-1}} (\mathbf{x}_1 - \boldsymbol{\mu}, \mathbf{x}_2 - \boldsymbol{\mu}, \dots, \mathbf{x}_m - \boldsymbol{\mu})$ then $\mathbf{C} = \mathbf{X}\mathbf{X}^T$. If $\mathbf{v}$ is an eigenvector of $\mathbf{C}$ so that $\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$, then multiply on the left by $\mathbf{v}^T$ we find

$$\mathbf{v}^T \mathbf{C} \mathbf{v} = \mathbf{v}^T \mathbf{X} \mathbf{X}^T \mathbf{v} = \lambda \mathbf{v}^T \mathbf{v}.$$

Defining $\mathbf{u} = \mathbf{X}^T \mathbf{v}$ then

$$\lambda = \frac{\mathbf{u}^T \mathbf{u}}{\mathbf{v}^T \mathbf{v}} = \frac{\|\mathbf{u}\|^2}{\|\mathbf{v}\|^2} \geq 0.$$

- (d) Prove that the set of positive semi-definite matrices forms a convex set. [5 marks]

For any two PSD matrices $M_1, M_2 \succeq 0$ the convex combination $M_3 = aM_1 + (1-a)M_2$ (where $a \in [0, 1]$) has a quadratic form with an arbitrary vector v of $v^T M_3 v = a v^T M_1 v + (1-a) v^T M_2 v$. But as $M_1, M_2 \succeq 0$ then $v^T M_1 v \geq 0$ and $v^T M_2 v \geq 0$. Since $a, (1-a) \geq 0$ it follows that $v^T M_3 v \geq 0$ or $M_3 \succeq 0$. Thus all convex combinations of positive semi-definite matrices are positive semi-definite so they form a convex set.

- (e) In gradient descent (in one dimension) we would update the weight by

$$w^{(t+1)} = w^{(t)} - r L'(w^{(t)})$$

assume that $L(w) = w^2$, write down a closed form expression for $w^{(t)}$ in terms of the initial state $w^{(0)}$ and explain what happens when $r > 1$. [5 marks]

The update equation becomes

$$w^{(t+1)} = w^{(t)} - 2r w^{(t)} = (1 - 2r)w^{(t)}$$

thus

$$w^{(t)} = (1 - 2r)^t w^0.$$

and for $r > 1$ (and provided $w^{(0)} \neq 0$) $w^{(t)}$ will diverge exponentially fast (oscillating between positive and negative values).

- (f) For the linear programming problem of maximising $c^T x$ subject to the constraints $Mx = b$ write down a Lagrangian $L(x, \alpha)$ where α is a set of Lagrange multipliers. By rewriting the Lagrangian in its dual form provide an equivalent minimisation problem for the Lagrange multipliers.

[5 marks]

The Lagrangian is given by

$$L(x, \alpha) = c^T x - \alpha^T (Mx - b).$$

The solution to the constrained optimisation is equivalent to maximising the Lagrangian with respect to x at the same time as minimising it with respect to α . The Lagrangian can be rewritten as

$$L(x, \alpha) = b^T \alpha - x^T (M\alpha - c).$$

But this can be interpreted as choosing α to minimise $b^T \alpha$, subject to the constraints $M\alpha = c$.

- (g) Show that the beta distribution $\text{Beta}(p|a, b) = p^{a-1} (1-p)^{b-1} / B(a, b)$ is a conjugate prior to the binomial likelihood $\text{Binom}(k|n, p) = \binom{n}{k} p^k (1-p)^{n-k}$. Derive update equations for the parameters of the posterior distribution after observing k successes and $n - k$ failures. [5 marks]

We only need to consider the functional form with respect to p . Thus the posterior is proportional to

$$f(p|k) \propto p^k (1-p)^{n-k} \times p^{a-1} (1-p)^{b-1} \propto \text{Beta}(p|a+k, b+n-k)$$

The updated equation is thus $(a, b) \rightarrow (a+k, b+n-k)$.

- (h) Show that if X and Y are independent random variables then the entropy, $H_{(X,Y)}$ for the two random variables is equal to the sum of the entropies for X and Y . [5 marks]

By definition

$$H_{(X,Y)} = - \sum_{X,Y} \mathbb{P}(X, Y) \log(\mathbb{P}(X, Y))$$

but if X and Y are independent then $\mathbb{P}(X, Y) = \mathbb{P}(X) \mathbb{P}(Y)$. Thus $\log(\mathbb{P}(X, Y)) = \log(\mathbb{P}(X)) + \log(\mathbb{P}(Y))$ and

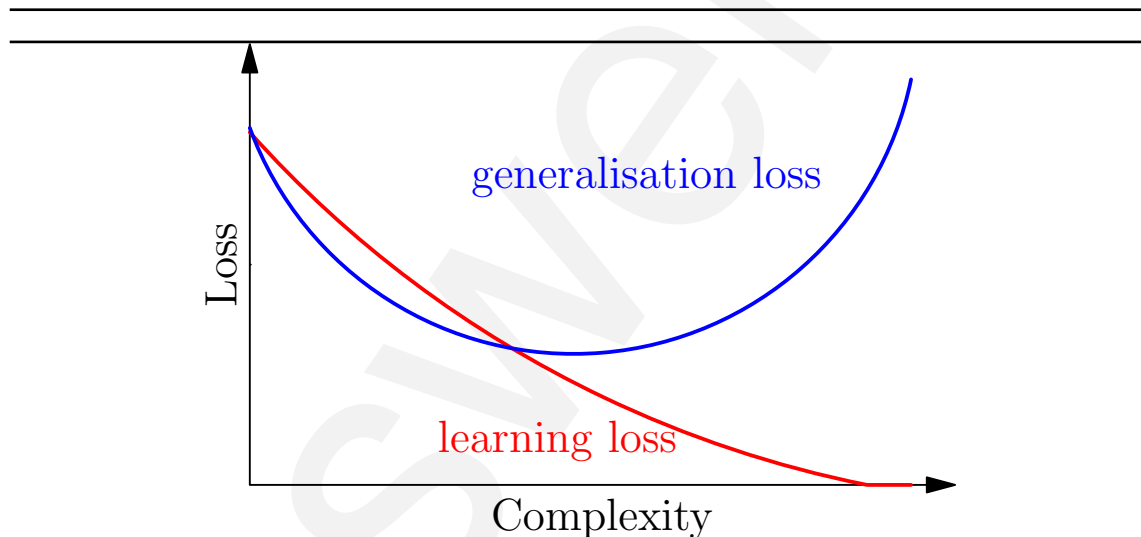
$$\begin{aligned} H_{(X,Y)} &= - \sum_{X,Y} \mathbb{P}(X) \mathbb{P}(Y) (\log(\mathbb{P}(X)) + \log(\mathbb{P}(Y))) \\ &= - \sum_X \mathbb{P}(X) \log(\mathbb{P}(X)) - \sum_Y \mathbb{P}(Y) \log(\mathbb{P}(Y)) = H_X + H_Y. \end{aligned}$$

End of question A1

Section B

B 2

- (a) On the axes given sketch the typical behaviour of the training and generalisation errors as a function of the models complexity (e.g the number of parameter). [5 marks]



- (b) Explain the training and generalisation loss in terms of the bias-variance dilemma. [5 marks]

As you increase the complexity of the model the learning machine is able to fit the training data (resulting in a continuous reduction in the training loss). This corresponds to a reduction in the bias resulting in an initial decrease in the generalisation error. However, as the complexity of the model grows the sensitivity of the model to the data usually increases giving an increase in the variance which starts to dominate the reduction in the bias resulting in the generalisation error increasing as the model becomes too complex.

- (c) Explain how in the Bayesian framework the model avoids overfitting. [5 marks]

In the Bayesian framework we compute the probability for all models. By averaging over all models we avoid having to choose the one that fits the data the best. Typically, by choosing the best fitting model we choose one that by chance does well on the training set. Avoiding choosing the best model, but averaging over the large number of models that fit the training data reasonably stops overfitting.

- (d) Explain how hyper-parameter tuning can result in over-fitting, and how we can avoid this in Bayesian inference [5 marks]

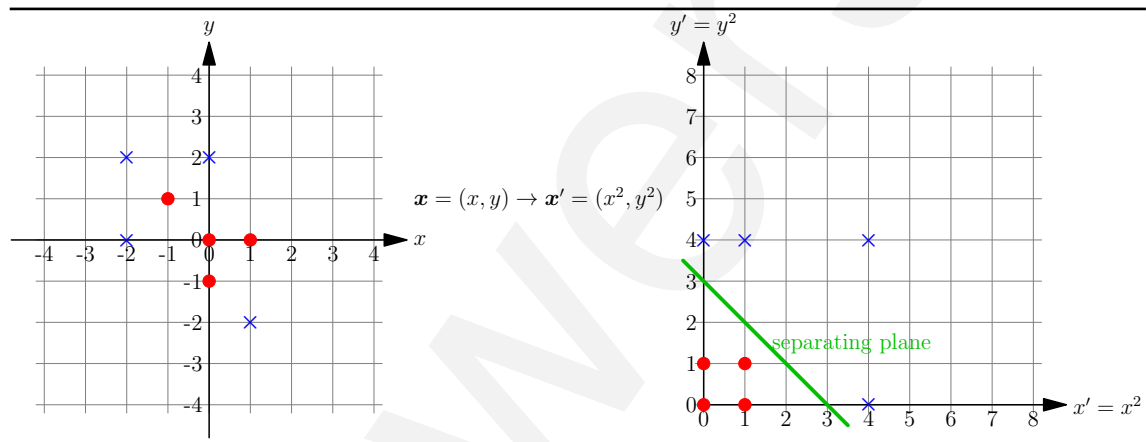
In hyper-parameter tuning if we use Maximum Likelihood to choose the hyper-parameters (so called, ML-II) then we are again choosing a model

based on the training data which is prone to overfit by choosing a model that by chance fits the finite training data set. To avoid this we can put a probability distribution on the hyper-parameters and use Bayesian inference to obtain a posterior distribution. We would then need to marginalise out the hyper-parameters using this posterior distribution.

End of question B2

B 3

- (a) Show how the data points $\{x_i = (x_i, y_i) | i = 1, 2, \dots\}$ shown below transform under the mapping $x = (x_i, y_i) \rightarrow x' = (x_i^2, y_i^2)$ and sketch the position of the maximal margin dividing plane in the new feature space. [5 marks]



- (b) Explain the kernel trick. [5 marks]

The kernel trick allows us to map data into a high dimensional feature space $x \rightarrow \phi(x)$. This can be carried out for sufficiently simple machines where parameter optimisation involve the dot product $\phi^T(x)\phi(y)$. This defines a kernel function $K(x, y) = \phi^T(x)\phi(y)$. Provided the kernel function is positive semi-definite, this decomposition is always possible. In fact, we never need to explicitly calculate the extended features $\phi(x)$. This often makes working in the extended feature space very efficient as $K(x, y)$ may be quick to calculate.

- (c) Define what it means for a kernel to be positive semi-definite (giving different properties a positive semi-definite kernel satisfies). Explain why it is necessary for the kernel function of an SVM to be positive semi-definite. [5 marks]

A positive semi-definite kernel satisfies a number of different conditions

- (i) All its eigen-functions have non-negative eigenvalues
 (ii) The quadratic form is non-negative. That is for all functions $f(x)$

$$\int \int f(x) K(x, y) f(y) dx dy \geq 0.$$

- (iii) Any positive semi-definite kernel can be written as a dot (inner) product between two vector functions

$$K(x, y) = \phi^T(x) \phi(y).$$

It is important for a kernel to be positive semi-definite in order that the condition of the margin being positive is meaningful. That is, if the eigenvalues were negative, then the “distance” would not necessarily be non-negative.

- (d) Using the fact that any positive semi-definite kernel can be decomposed as

$$K(x, y) = \sum_i \phi_i(x) \phi_i(y)$$

show that the product of two kernel functions is positive semi-definite. [5 marks]

- If $K_1(x, y)$ and $K_2(x, y)$ are valid kernels then so is $K_3(x, y) = K_1(x, y) K_2(x, y)$
- Writing $K_1(x, y) = \sum_i \phi_i^1(x) \phi_i^1(y)$ and $K_2(x, y) = \sum_j \phi_j^2(x) \phi_j^2(y)$ then

$$K_3(x, y) = \sum_{i,j} \phi_{ij}^3(x) \phi_{ij}^3(y)$$

where $\phi_{ij}^3(x) = \phi_i^1(x) \phi_j^2(x)$. In other words,

$$\int f(x) K_3(x, y) f(y) dx dy = \sum_{i,j} \left(\int \phi_i^1(x) \phi_j^2(x) f(x) dx \right)^2 \geq 0.$$

End of question B3

B 4

- (a) Explain why Monte Carlo techniques are often used to solve Bayesian inference problems. [10 marks]

For many Bayesian problems the posterior has no closed form solution so it cannot be expressed. Monte Carlo techniques allow us to draw samples from the posterior which allow us to compute many quantities of interest such as the posterior mean (often the best single prediction) and the posterior variance (giving an indication of the expected error).

- (b) Briefly describe in words the use of the MCMC algorithm in Bayesian inference. [10 marks]

In MCMC we explore the posterior distributions by making small jumps in the possible solutions with a probability that satisfies detail balance. That is, the probability of making a move is dependent on the ratio of the posterior probability of the solution before and after the move. This ensures that the distributions of sampled points converges to the posterior distribution. We need to throw away the initial points (burn-in phase). This gives us sample points from the posterior (although to be independent we need to wait for the points to decorrelate). The advantage of MCMC for Bayesian inference is that we can sample from the posterior without having to compute the marginal likelihood $P(\mathcal{D})$, which is usually costly and often intractable.

End of question B4

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